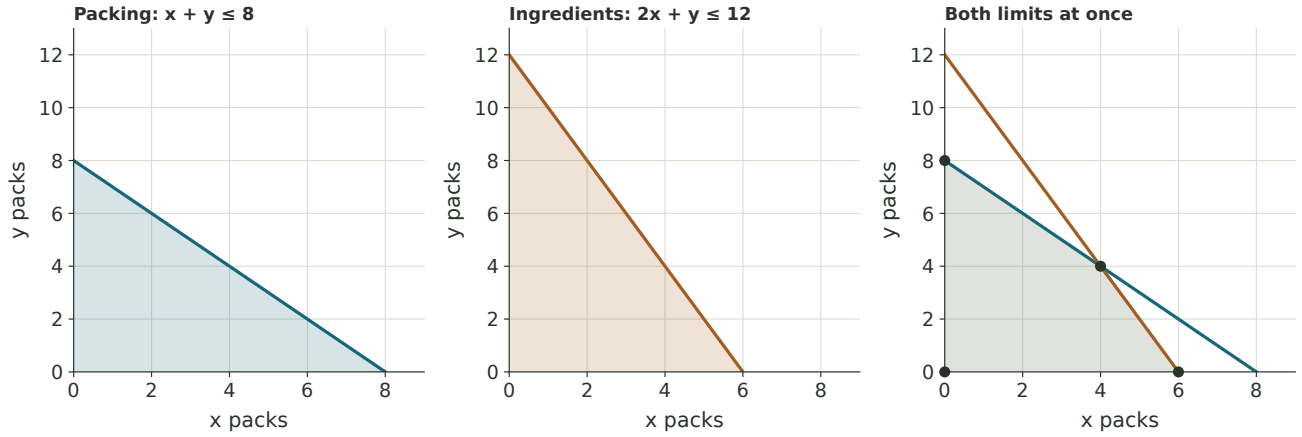
WEEK 8 / PART I / CONVEX GEOMETRY

# What is possible?

Turn resource limits into a picture of allowable plans. Learn what a point means, what a boundary means, and why every condition must hold.




**The setting for the week**

A school club makes two kinds of snack packs. It has eight packing slots and twelve ingredient portions. Type A uses twice as many portions as type B. Which complete production plans can it actually carry out?

Week 7 compared costs, revenue, and profit. Here we add limits. Use multiplication, fractions, coordinate pairs, and simple equations to describe what is allowed before choosing what is best. No previous study of linear programming is assumed.

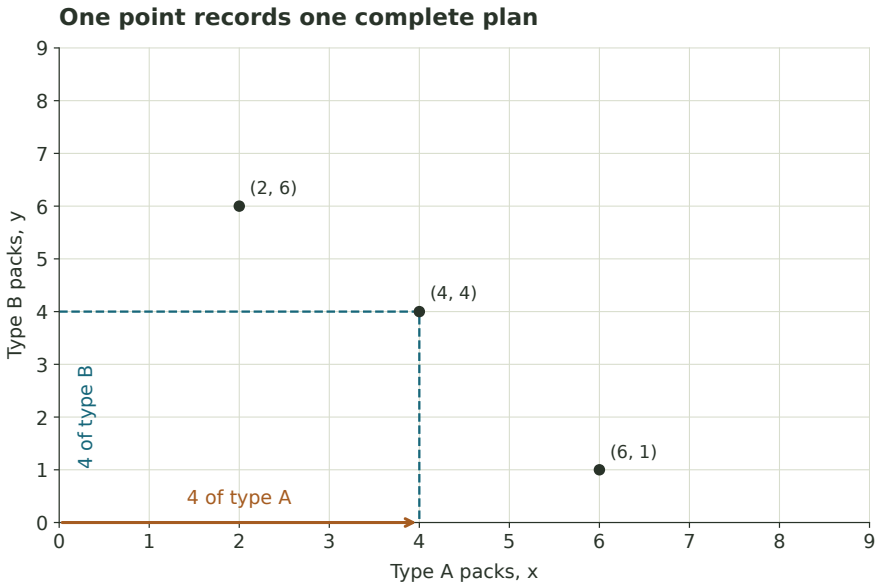
Each resource constraint defines a set of possible plans. Their intersection is the feasible region. Its boundaries and vertices reveal how the resource limits act together.

Complete online reading · Feasible-region explorer · Guided laboratory

## 8.1 One point records one complete plan

Let $x$ be the number of full tune-ups and $y$ the number of quick repairs. These are *decision variables*: the quantities the shop chooses. A schedule $(4, 4)$ means four tune-ups and four quick repairs. The first coordinate counts tune-ups; the second counts quick repairs.

On the graph, move horizontally to the x-value, then vertically to the y-value. The point is a whole production plan, not a moment in time. Later, the line between two schedules will help us understand averages of possible workdays.




Read (2,6) as "2 tune-ups and 6 quick repairs." This is a graph of choices. Neither axis measures time, and a higher point is not automatically a better plan.

| Per bicycle or resource | Tune-up, x | Quick repair, y | Available |
|---|---|---|---|
| Bicycles accepted | 1 | 1 | 8 bikes |
| Mechanic hours | 2 | 1 | 12 hours |
| Contribution after parts | \$30 | \$20 | What the shop wants to increase |

Here, *contribution* means money remaining after buying the replacement parts. A real shop also has wages, rent, and other expenses. We use simple amounts so that the different schedules are easy to compare. Subtracting the same fixed daily expense from every schedule would not change which one has the greatest contribution.

**The graph and the real workday**

For the geometry, the shaded region includes points with fractional coordinates. A real shop cannot finish half a bicycle, so we will check that a final schedule uses whole numbers of jobs.

**Pause and try 1**

What does (5,2) mean? What would (2,5) mean, and why is it a different decision?

## 8.2 Each constraint compares a total with a limit

Each repair involves one bicycle. Since the shop can accept at most eight bikes, the total number of repairs must satisfy $x + y \leq 8$. The symbol $\leq$ means "less than or equal to": eight are fine, but nine would exceed the limit.

A full tune-up takes two mechanic-hours, and a quick repair takes one. That gives $2x + y$ hours of work, so $2x + y \leq 12$. The coefficient 2 means *two hours per tune-up*. Multiplying hours per job by the number of jobs gives total hours.

$$x + y \leq 8, \qquad 2x + y \leq 12, \qquad x \geq 0, \quad y \geq 0.$$

The last two constraints rule out negative numbers of repairs. A plan is *feasible* if it satisfies *every* constraint. Every resource limit must be satisfied at the same time.

**Example · Test (5,3) and (3,3)**

Five tune-ups and three quick repairs involve $5 + 3 = 8$ bikes, which fits the capacity. But the schedule requires $2(5) + 3 = 13$ mechanic-hours, one more than the shop has. It is *infeasible* even though the number of bikes is acceptable.
Three tune-ups and three quick repairs involve six bikes and nine mechanic-hours. This is a feasible schedule, leaving room for two more bikes and three hours of mechanic time. The unused amounts are called *slack*.

| Words | Mathematical form | Meaning |
|---|---|---|
| At most 8 | total $\leq 8$ | 8 is allowed; larger totals are not |
| At least 8 | total $\geq 8$ | 8 is allowed; smaller totals are not |
| Exactly 8 | total $= 8$ | Only 8 is allowed |
| Less than 8 | total $< 8$ | 8 itself is excluded |

A constraint is *binding*, or active, at a plan when it holds with equality. At (4,4), both resource limits bind: $4 + 4 = 8$ and $2(4) + 4 = 12$. A binding constraint has no slack at that plan.

**Pause and try 2**

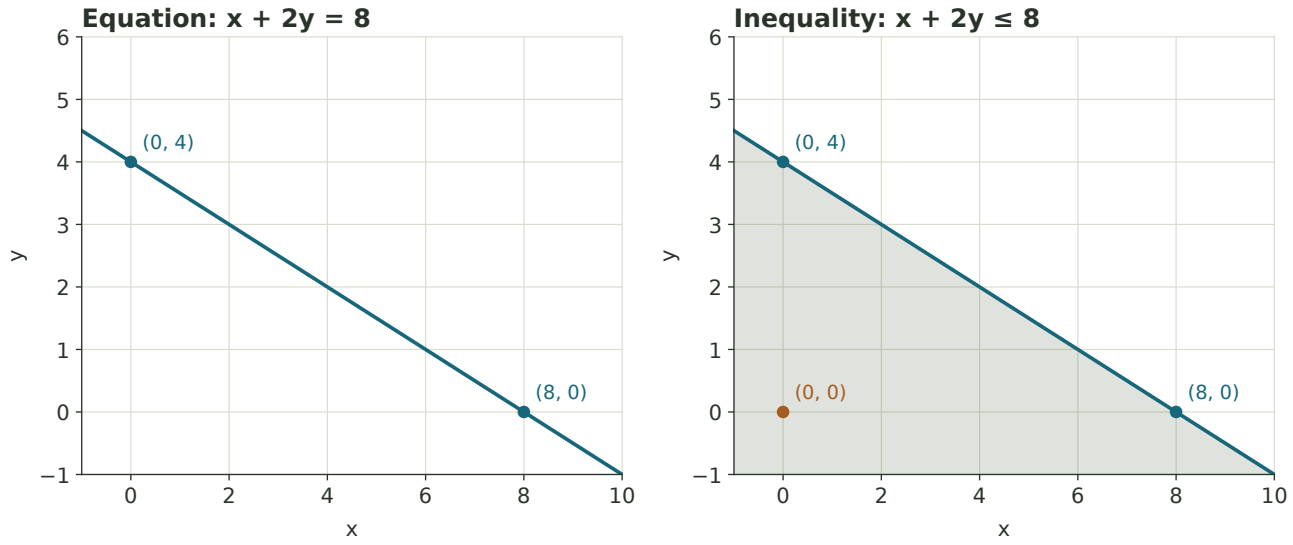
Could the shop complete six tune-ups and one quick repair? Show both calculations.

## 8.3 The boundary is a line; the inequality permits a whole side

Here is another situation. A community center has eight room-hours available. If $x$ counts one-hour activities and $y$ counts two-hour workshops, the schedule must satisfy $x + 2y \leq 8$. The *boundary line* $x + 2y = 8$ represents schedules that use all eight hours; the inequality also allows schedules that use fewer.

**Example · Find two intercepts, then test one point**

1. Set $y = 0$: then $x = 8$, giving the x-intercept (8,0).
2. Set $x = 0$: then $2y = 8$, so $y = 4$, giving the y-intercept (0,4).
3. Draw the line through those points.
4. Test a point off the line. At (0,0), $0 + 2(0) = 0 \leq 8$, so the side containing the origin is permitted.




The equation describes a line. The inequality describes a half-plane: that line together with all points on one side. The panels show only a window of the plane; a half-plane continues beyond the picture.

For $\leq$ or $\geq$, draw the boundary as a solid line because equality is allowed. For a strict inequality, such as $<$, use a dashed boundary to show exclusion. Our optimization examples use included boundaries. If the origin lies on the boundary, choose a different test point that does not.

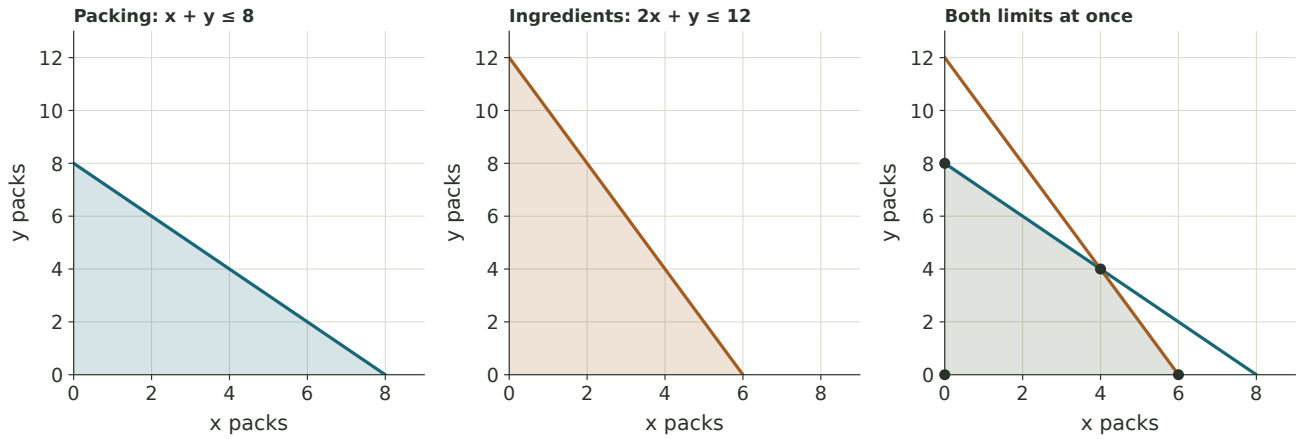
Back at the bike shop, $x \geq 0$ and $y \geq 0$ restrict us to the *first quadrant including its axes*. This restriction does not make a single half-plane a bounded region on its own; we must inspect all the conditions together.

**Pause and try 3**

For x + 2y $\leq$ 8, is (2,3) on the boundary, strictly on the permitted side, or outside? What about (4,3)?

## 8.4 The feasible region is the overlap

Return to the bike shop's limits, $x + y \leq 8$ and $2x + y \leq 12$, with nonnegative coordinates. Each limit permits its own side of a boundary. The *feasible region* is their *intersection*: the points common to all the permitted sets.




The first two panels show one resource limit at a time, with nonnegative coordinates. Keep a plan only if it is permitted in both panels. The final shaded polygon is the feasible region.

A *vertex* (plural: vertices) is a corner of this region. Corners come from intersecting boundaries, including the axes. A boundary intersection is only a *candidate* vertex until we verify all constraints.

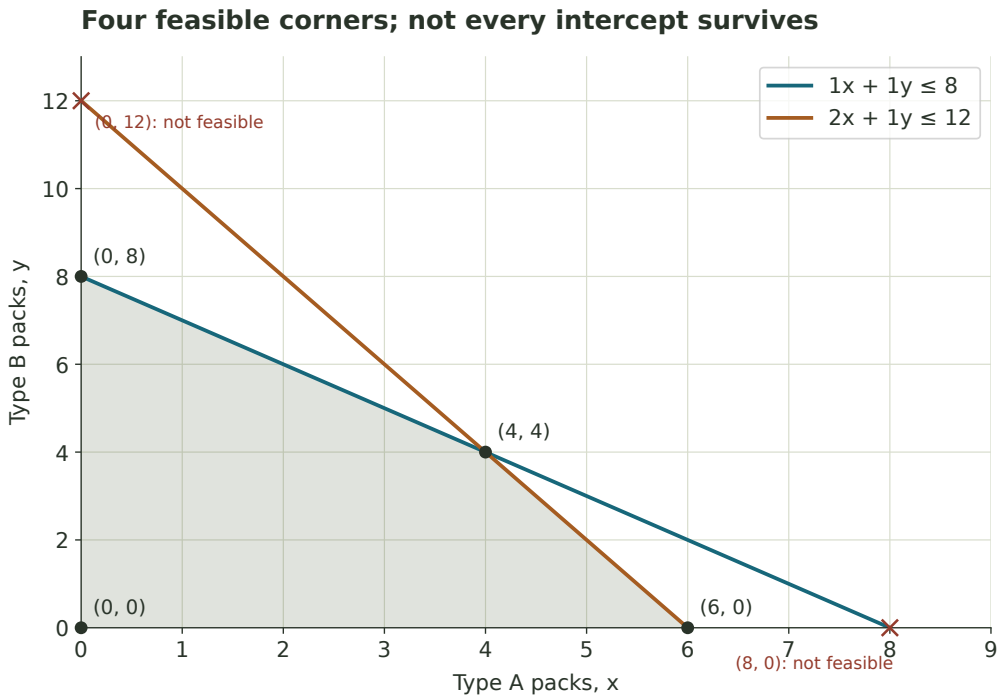
**Example · Find the non-axis corner**

Solve the two resource boundaries simultaneously:

$$x + y = 8, \qquad 2x + y = 12.$$

Subtract the first equation from the second: $(2x + y) - (x + y) = 12 - 8$, so $x = 4$. Substitute into $x + y = 8$ to get $y = 4$. Thus the boundaries meet at (4,4). Both coordinates are nonnegative, so this intersection is feasible.

On the x-axis, $y = 0$. The limits require both $x \leq 8$ and $x \leq 6$, so the tighter upper limit is 6. The x-axis corner is (6,0). On the y-axis, the tighter limit is 8, giving (0,8). The origin is the fourth corner.




Two different ideas: (8,0) is an intercept of the bike-capacity boundary, but it is not a feasible corner because it needs 16 mechanic-hour portions. Likewise, (0,12) violates the bike limit.

| Corner | Bikes x+y ≤ 8 | Hours 2x+y ≤ 12 | Binding resource limits |
|---|---|---|---|
| (0,0) | 0 | 0 | Neither |
| (6,0) | 6 | 12 | Hours |
| (4,4) | 8 | 12 | Both |
| (0,8) | 8 | 8 | Bikes |

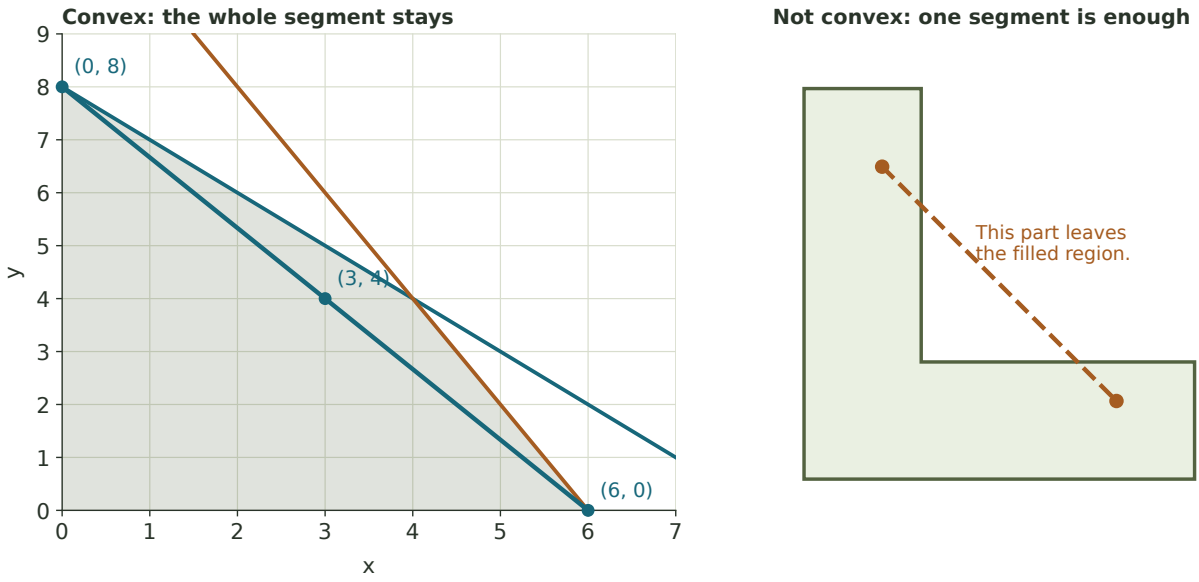
The axes are also boundaries: at (6,0), for example, the nonnegativity condition $y \geq 0$ binds along with the mechanic-hour limit. A corner does not always use up both resources.

**Pause and try 4**

For x+y≤7 and 2x+y≤10 with x,y≥0, find the intersection of the two non-axis boundaries. Is (7,0) a feasible corner?

## 8.5 Convexity means that the whole segment stays inside

A set is *convex* if the straight segment between *any two* points in the set remains entirely in the set. A filled triangle, filled rectangle, and filled disk are convex. A ring is not: a segment across the hole leaves the set. The circumference of a circle alone is also not convex; the disk includes the interior.




To establish nonconvexity, one counterexample segment is enough. Convexity requires every such segment to remain inside, not just one convenient segment.

### Example · Average two feasible plans

The halfway plan between (6,0) and (0,8) is $((6+0)/2, (0+8)/2) = (3, 4)$. It involves seven bikes and ten mechanic-hours, both within the daily limits.
Why does averaging work for any two feasible real-valued plans? Each resource total at the average plan is the average of the two resource totals. If each total is at most a limit, their average is too. The same reasoning works for any mixture with nonnegative weights adding to 1.

Every half-plane is convex, and intersections of convex sets are convex. This explains the shape of our feasible region. A *union* is different: two separated filled rectangles are individually convex, but taking all points in either rectangle produces a set with a gap.

Averages are different from actual workdays. Halfway between (1,0) and (0,1) is (0.5,0.5), which can describe an average over several days but not a day's schedule of whole bicycles.

### Pause and try 5

Why is a filled rectangle convex while two separated filled rectangles, taken together, are not?

## Part I Review

Use nonnegative real coordinates unless the question explicitly asks for whole numbers. Show a substitution, boundary calculation, or segment argument.

1. Define variables for a workshop making chairs and tables. What does the plan $(5, 2)$ mean? Explain why $(2, 5)$ is different.
2. A type-A item uses 3 labor-hours and a type-B item 2. Write a 60-hour upper limit with units. Test $(10, 10)$ and $(15, 8)$ and report slack or the shortfall.
3. For $x + 2y \leq 8$, find both boundary intercepts. Identify the permitted side by testing a point off the boundary. Classify $(2, 3)$ and $(4, 3)$.
4. Find every feasible corner for $x + y \leq 7$, $2x + y \leq 10$, $x, y \geq 0$. Show the intersection calculation. Why is $(7, 0)$ not a corner of the feasible region?
5. Find the corners for $3x + 2y \leq 120$, $x + 2y \leq 60$, $x, y \geq 0$. Test $(20, 20)$ and $(25, 20)$ against both resource limits.
6. For the workshop constraints $2x + 4y \leq 40$, $x + y \leq 14$, find the four corners. Check proposed plans $(8, 5)$ and $(10, 5)$.
7. Which sets are convex: a filled disk, a circle's circumference alone, a filled rectangle, a ring, a line segment? Explain a counterexample segment for each nonconvex set.
8. Why is an intersection of convex sets convex? Give a counterexample showing that a union need not be convex.
9. For the club's region, average $(6, 0)$ and $(0, 8)$. Check the resulting plan against $x + y \leq 8$ and $2x + y \leq 12$. Why need a midpoint of two whole-number plans not be a whole-number plan?
10. Keep the club's packing limit at 8 and raise the ingredient allowance from 12 to 16. Do old feasible plans remain feasible? Give one newly feasible plan.
11. If $x + y \leq 8$ is already imposed, does adding $x + y \leq 10$ remove any choices? What does it mean for a constraint to be redundant?
12. Describe the feasible sets for (a) $x + y \leq 4$ and $x + y \geq 6$; (b) $x + y \leq 0$ with $x, y \geq 0$. Why should we not assume every system creates a filled four-sided polygon?

## Solutions and Reasoning

**Short check 1.** The first schedule has five tune-ups and two quick repairs; the second has two tune-ups and five quick repairs. They use different amounts of mechanic time and earn different amounts.

**Short check 2.** The seven bikes fit the daily bike limit, since $6+1 = 7 \leq 8$. But the work takes thirteen hours, since $2(6)+1 = 13 > 12$. The shop cannot finish that schedule in the time available.

**Short check 3.** At (2,3), 2+2(3)=8, so it is on the included boundary. At (4,3), the total is 10, so it is outside. An included boundary point is feasible for that inequality.

**Short check 4.** Subtracting gives x=3, then y=4, so the intersection is (3,4). The intercept (7,0) fails $2x+y \leq 10$ because it uses 14. The feasible x-axis corner is (5,0).

**Short check 5.** A segment between any two points in one filled rectangle stays in that rectangle. With two separated rectangles, choose one point in each: the connecting segment crosses a gap outside their union.

### Part I Review: solutions

1. Let $x$ count chairs per week and $y$ tables per week. $(5, 2)$ means five chairs and two tables. Switching the coordinates switches the products, which may change resource use and value.

2. $3x + 2y \leq 60$ hours, where the coefficients are hours per item. $(10, 10)$ uses $30 + 20 = 50$ hours, leaving 10. $(15, 8)$ uses $45 + 16 = 61$ hours and exceeds the allowance by 1.

3. Intercepts are $(8, 0)$ and $(0, 4)$. The origin gives $0 \leq 8$, so its side is permitted. $(2, 3)$ is on the included boundary; $(4, 3)$ is outside because its total is 10.

4. Subtract $x + y = 7$ from $2x + y = 10$: $x = 3$, then $y = 4$. Corners are $(0, 0), (5, 0), (3, 4), (0, 7)$. $(7, 0)$ uses 14 under the second limit, exceeding 10.

5. Subtract the resource equations to get $2x = 60$, so $x = 30, y = 15$. Corners are $(0, 0)$, $(40, 0)$, $(30, 15)$, and $(0, 30)$. At $(20, 20)$, use is 100 and 60: feasible. At $(25, 20)$, use is 115 and 65: the second limit fails.

6. Divide the first boundary by 2 and subtract $x + y = 14$ from $x + 2y = 20$, obtaining $y = 6, x = 8$. Corners are $(0, 0), (14, 0), (8, 6), (0, 10)$. $(8, 5)$ uses 36 wood units and 13 slots: feasible. $(10, 5)$ uses 40 wood units but 15 slots: infeasible.

7. The filled disk, filled rectangle, and line segment are convex. Opposite points of a circle's circumference have a segment through the missing interior. Opposite sides of a ring have a segment through the hole. These counterexamples establish nonconvexity.

8. If two points lie in every set, their joining segment lies in every convex set and thus in the intersection. For a union, take two separated filled rectangles and join a point of one to a point of the other: the segment crosses a gap.

9. The midpoint is $(3, 4)$, using 7 slots and 10 ingredient portions. It is feasible. But the midpoint of whole-number points $(1, 0)$ and $(0, 1)$ is $(0.5, 0.5)$, which is not a whole-number plan.

10. Every old feasible plan remains feasible because the ingredient upper limit is relaxed. $(8, 0)$ is newly feasible: it uses 8 slots and 16 portions.

11. No choices are removed: a total at most 8 is automatically at most 10. A redundant constraint adds no restriction beyond those already imposed.

12. (a) The set is empty: one total cannot be at most 4 and at least 6. (b) Only $(0, 0)$ is feasible. Other systems can produce triangles, larger polygons, line segments, or unbounded sets. The shape must be established from the actual conditions.

## Continue with Part II

Now add an objective to the feasible region. The online optimizer and Part II explain which conditions justify a corner comparison, what ties mean, and why the best continuous plan may not be a usable whole-number plan.